An improved Canny edge detection algorithm for color image

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***Abstract*—The traditional Canny edge detection method is widely used in gray image processing. However, this traditional algorithm is unable to deal with color images and the parameters in the algorithm are difficult to be determined adaptively. In this paper, an improved Canny algorithm is proposed to detect edges in color image. The proposed algorithm is composed of the following steps: quaternion weighted average filter, vector Sobel gradient computation, non-maxima suppression based on interpolation, edge detection and connection. Experimental results show that the proposed algorithm outperforms other color image edge detection methods and can be widely used in color image processing.**

***Keywords*—edge detection, color image, vector Sobel operator, non-maxima suppression, Canny algorithm**

I. INTRODUCTION

The widespread applications of color images, such as image recognition, video surveillance and medical image analysis, have led to an increasing interest in color image processing. However, the research of color image processing is obviously later than gray image. Thus it is necessary to do more research on this area.

Image segmentation is an important work in the field of image processing. It is the preprocessing step for image analysis and recognition. There are many kinds of segmentation methods, such as histogram segmentation, edge detection, region fusion and so on. Among these methods, edge detection is a direct way to acquire the outline of the interesting object [1].

Many researches have been done in the filed of gray image edge detection. However, color image edge detection algorithm is not studied adequately. Nowadays, there are two easy ways to segment color image. The first way is to change the color image to a gray one and process it as a gray image. The other is to execute the gray image edge detection algorithm separately in three channels of color image and fuse the three channels results by a rule [2]. Usually, there are two kinds of fusion rules. One is to determine the edges by logic rules. For example, if there are edge pixels in two channels for a color vector point, this location is to be determined as edge pixel. The other way is to calculate the average, maximum or mean square root of the three channels and then determine the edges by a threshold.

For gray images, there are many edge detection algorithms such as Robert operator, Sobel operator, Kirsch operator and so on (see Fig.1). These operators are designed to detect the step response due to the edge location is the discontinuous part of the image. These operators are widely used in edge detection because of the easy computations. However, these operators are sensitive to noise and the edge determined is discontinuous.

-5	-5	-5
3	0	3
3	3	3

3	-5	-5
3	0	-5
3	3	3

3	3	-5
3	0	-5
3	3	-5

3	3	3
3	0	-5
3	-5	-5

3	3	3
3	0	3
-5	-5	-5

3	3	3
-5	0	3
-5	-5	3

-5	3	3
-5	0	3
-5	3	3

-5	-5	3
-5	0	3
3	3	3

Figure 1. Kirsch operators with 3×3 size and eight directions

Canny algorithm is another method for edge detection, which has the advantages of good SNR (Signal-to-Nose Ratio), accurate location and edge thinning [3]. However, the traditional Canny algorithm is used in gray image. The research on how to expand this algorithm to color image is necessary and useful. Thus, this paper presents an improved color image Canny algorithm based on the vector concept. Comparison and application experiments are carried out to estimate the proposed algorithm.

II. THE TRADITIONAL CANNY ALGORITHM

The traditional Canny algorithm is used in gray images and can be divided into four parts: image smoothing, gradient computation, non-maxima suppression, edge determination and connection.

A. Image Smoothing

Image smoothing is a necessary step to remove noise before edge detection. The traditional Canny algorithm uses two orders Gaussian function to smooth image. The function is described as [4]:

$$G(x,y)=\frac{1}{2\pi\sigma^2}\exp[-\frac{x^2+y^2}{2\sigma^2}]. \quad (1)$$

The gradient vector is:

$$\nabla\mathbf{G}=[\frac{\partial G}{\partial x},\frac{\partial G}{\partial y}]^T. \quad (2)$$

The two-dimensional Gaussian filter can be divided into two row and column filters as:

$$\frac{\partial G}{\partial x}=kx\cdot\exp|-\frac{x^2}{2\sigma^2}|\cdot\exp|-\frac{y^2}{2\sigma^2}|. \quad (3)$$

$$\frac{\partial G}{\partial y}=ky\cdot\exp|-\frac{y^2}{2\sigma^2}|\cdot\exp|-\frac{x^2}{2\sigma^2}|. \quad (4)$$

σ is a parameter which controls the smoothing degree and k is a constant. When σ is small, the smoothing ability is low but the edge preservation ability is high. Whereas, when σ is large, the situation is contrary.

B. Gradient Computation

The one order finite difference is applied to do gradient computation as follows:

$$E^x_{i,j}=\frac{1}{2}(I'_{i,j+1}-I'_{i,j}+I'_{i+1,j+1}-I'_{i+1,j}). \quad (5)$$

$$E^y_{i,j}=\frac{1}{2}(I'_{i,j}-I'_{i+1,j}+I'_{i,j+1}-I'_{i+1,j+1}). \quad (6)$$

The gradient amplitude is defined as:

$$M_{i,j}=\sqrt{(E^x_{i,j})^2+(E^y_{i,j})^2}. \quad (7)$$

The gradient direction is defined as:

$$\theta_{i,j}=\tan^{-1}|\frac{E^y_{i,j}}{E^x_{i,j}}|. \quad (8)$$

C. Non-maxima Suppression

The purpose of NMS (non-maxima suppression) is to thin the edges. In the traditional Canny algorithm, the gradient value $M_{i,j}$ of object pixel, which is to be processed by NMS, is compared with two adjacent pixels along the gradient direction. If $M_{i,j}$ is larger than the two gradient values of the adjacent pixels, $M_{i,j}$ is kept unchanged; otherwise, $M_{i,j}=0$.

D. Edge Determination and Connection

After the process of NMS, two thresholds are used to determine real edges. Usually, the high threshold is applied to acquire the main outline, which contains fewer fake edges but the edge may be discontinuous. While, the low threshold is applied to determine more edges, which may contain more fake edges. Thus, the low threshold is used to connect the discontinuous part of the outline determined by the high threshold.

E. Disadvantages of the Traditional Canny Algorithm

As aforementioned, the traditional Canny algorithm is unable to deal with color image. Wenliang Du proposed a method to expand Canny algorithm to color image [5]. In this method, the color image is changed into gray one first. Then the traditional Canny algorithm is applied to process the gray image. Finally, the edges are marked in the color image. It is necessary to note that this method still belong to gray image processing. There are also other problems of the traditional Canny algorithm. One is how to reach a balance between noise removal and edge preservation. The other is how to determine the thresholds adaptively. Base on this analysis, a color Canny algorithm is proposed to resolve these problems.

III. Improved Canny algorithm

To deal with the weakness of traditional Canny algorithm, we involve the concept of quaternion weighted average filter and whole vector analysis. The improved algorithm is still be divided into four step as follows.

A. Quaternion Weighted Average Filter

The quaternions are the extension of two-dimensional complex numbers [6]. A quaternion number $\mathbf{q}$ is a four-dimensional number, which consists of one real part and three imaginary parts. It can be represented as the hypercomplex form [7, 8]:

$$\mathbf{q}=a+bi+cj+dk, \quad (9)$$

where a, b, c and d are real; i, j and k are complex operators obeying the following rules:

$$\begin{aligned} &i^2=j^2=k^2=ijk=-1,\\ &ij=k,\ jk=i,\ ki=j,\\ &ji=-k,\ kj=-i,\ ik=-j. \end{aligned} \quad (10)$$

When $a=0$, q is named as a pure quaternion. The modulus and conjugate of a quaternion are defined as follows respectively:

$$|\mathbf{q}|=\sqrt{a^2+b^2+c^2+d^2}, \quad (11)$$

$$\bar{\mathbf{q}}=a-bi-cj-dk. \quad (12)$$

Consider the widely used RGB color space, a color image can be represented as the pure quaternion form:

$$\mathbf{q}_{x,y}=r_{x,y}i+g_{x,y}j+b_{x,y}k, \quad (13)$$

where x, y denote the coordinates and $r,g,b\in\{0,1,2\ldots255\}$. In [9], a new color pixel distance d_q is defined which considers the intensity and color differences simultaneously. Based on this distance, we define QWAF (quaternion weighted average filter) as:

$$\mathbf{q}^{QWAF}_{x,y}=\frac{\sum_{i=1}^{n}w_i\mathbf{q}_i}{\sum_{i=1}^{n}w_i}, \quad (14)$$

where $i = 1, 2, \ldots, n$ stands for the pixels in filtering window; n is the size of window and selected as 9 in this paper. w_i is the weight and defined as:

$$w_i = e^{-d_q(\mathbf{q}_i, \mathbf{q}_5)}. \tag{15}$$

Note that $\mathbf{q}_5$ is the central pixel. When the distance between the pixel and window center is large, w_i is small; whereas, when the distance is small, w_i is large. Thus this filter can be used to remove Gaussian noise while preserve details.

B. Vector Sobel Operator

Sobel operator is a widely used method to compute gradient because of its easy format and noise insensitivity character. In gray image, the row and column Sobel operator are defined separately as follows:

$$\Delta H = \frac{1}{4}\begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}. \tag{16}$$

$$\Delta V = \frac{1}{4}\begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}. \tag{17}$$

A color pixel in RGB color space can be defined as a three-dimensional vector $\mathbf{I}$. For color image, we can define vector row and column Sobel operator as follows:

$$\Delta\mathbf{H} = (\mathbf{H}^- \quad 0 \quad \mathbf{H}^+). \tag{18}$$

$$\Delta\mathbf{V} = \begin{pmatrix} \mathbf{V}^- \\ 0 \\ \mathbf{V}^+ \end{pmatrix}. \tag{19}$$

$\mathbf{H}^+$, $\mathbf{H}^-$, $\mathbf{V}^+$ and $\mathbf{V}^-$ can be calculated as follows:

$$\mathbf{H}^+(x_0, y_0) = \frac{1}{4}[\mathbf{I}(x_0+1, y_0-1) + 2\mathbf{I}(x_0+1, y_0) + \mathbf{I}(x_0+1, y_0+1)]. \tag{20}$$

$$\mathbf{H}^-(x_0, y_0) = \frac{1}{4}[\mathbf{I}(x_0-1, y_0-1) + 2\mathbf{I}(x_0-1, y_0) + \mathbf{I}(x_0-1, y_0+1)]. \tag{21}$$

$$\mathbf{V}^+(x_0, y_0) = \frac{1}{4}[\mathbf{I}(x_0-1, y_0+1) + 2\mathbf{I}(x_0, y_0+1) + \mathbf{I}(x_0+1, y_0+1)]. \tag{22}$$

$$\mathbf{V}^-(x_0, y_0) = \frac{1}{4}[\mathbf{I}(x_0-1, y_0-1) + 2\mathbf{I}(x_0, y_0-1) + \mathbf{I}(x_0+1, y_0-1)]. \tag{23}$$

Where x_0, y_0 are the coordinates of a color pixel. The amplitude and direction can be defined as follows:

$$\begin{aligned} M(x_0, y_0) &= \sqrt{\|\Delta\mathbf{H}(x_0, y_0)\|^2 + \|\Delta\mathbf{V}(x_0, y_0)\|^2} \\ &= \sqrt{\|\mathbf{H}^+(x_0, y_0) - \mathbf{H}^-(x_0, y_0)\|^2 + \|\mathbf{V}^+(x_0, y_0) - \mathbf{V}^-(x_0, y_0)\|^2}. \end{aligned} \tag{24}$$

$$\theta = \arctan\left(\frac{\Delta V'(x_0, y_0)}{\Delta H'(x_0, y_0)}\right). \tag{25}$$

$\Delta V'$ and $\Delta H'$ can be computed as:

$$\Delta V'(x_0, y_0) = \begin{cases} \|\mathbf{V}(x_0, y_0)\| & When \ \|\mathbf{V}^+(x_0, y_0)\| \geq \|\mathbf{V}^-(x_0, y_0)\| \\ -\|\mathbf{V}(x_0, y_0)\| & Otherwise \end{cases}. \tag{26}$$

$$\Delta H'(x_0, y_0) = \begin{cases} \|\mathbf{H}(x_0, y_0)\| & When \ \|\mathbf{H}^+(x_0, y_0)\| \geq \|\mathbf{H}^-(x_0, y_0)\| \\ -\|\mathbf{H}(x_0, y_0)\| & Otherwise \end{cases}. \tag{27}$$

Where $\|\bullet\|$ is Euclidean 2-norm.

C. NMS based on Interpolation

The NMS based on interpolation is a method to compute the values of adjacent pixels using the interpolation of gradient amplitude (see Fig. 2) [10]. In Fig. 2, The direction of $I(x_0, y_0)$ is θ. The extended line along this direction intersects the 3×3 window at P_1 and P_2. d is the distance between P_1 and $I(x_0\text{-}1, y_0\text{+}1)$. Obviously, d=1-tanθ. $M(x_0-1, y_0+1)$ and $M(x_0, y_0+1)$ are defined separately as the gradient values of $I(x_0-1, y_0+1)$ and $I(x_0, y_0+1)$. M_{P_1} stands for the gradient of P_1 can be computed as:

$$M_{P_1} = dM(x_0-1, y_0+1) + (1-d)M(x_0, y_0+1). \tag{28}$$

In the same way, we can compute M_{P2} which is the gradient of P_2. Then compared the gradient of $I(x_0, y_0)$ with M_{P_1} and M_{P2} to suppress non-maxima value.

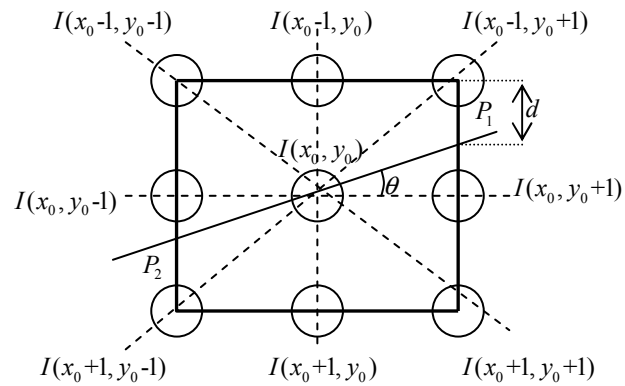


Figure 2. Non-maxima suppression based on interpolation

D. Threshold determination and edge connection

The method in [4] is a useful way to determine thresholds adaptively. Define $J_{\max}$ as the gradient value which occupies the most points in gradient histogram. $K_{\max}$ is the variance between $J_{\max}$ and all the other pixels in the image. $K_{\max}$ is computed as:

$$K_{\max} = \sqrt{\frac{\sum_{i=0}^{k} N_i \times (i - J_{\max})}{N}}. \tag{29}$$

Where k is the highest gradient value; N_i is the pixel number whose gradient value equals i; N is the total pixel number in the image. Then the high and low thresholds T_h and T_l are computed separately as:

$$T_h = J_{\max} + \beta \cdot K_{\max}. \tag{30}$$

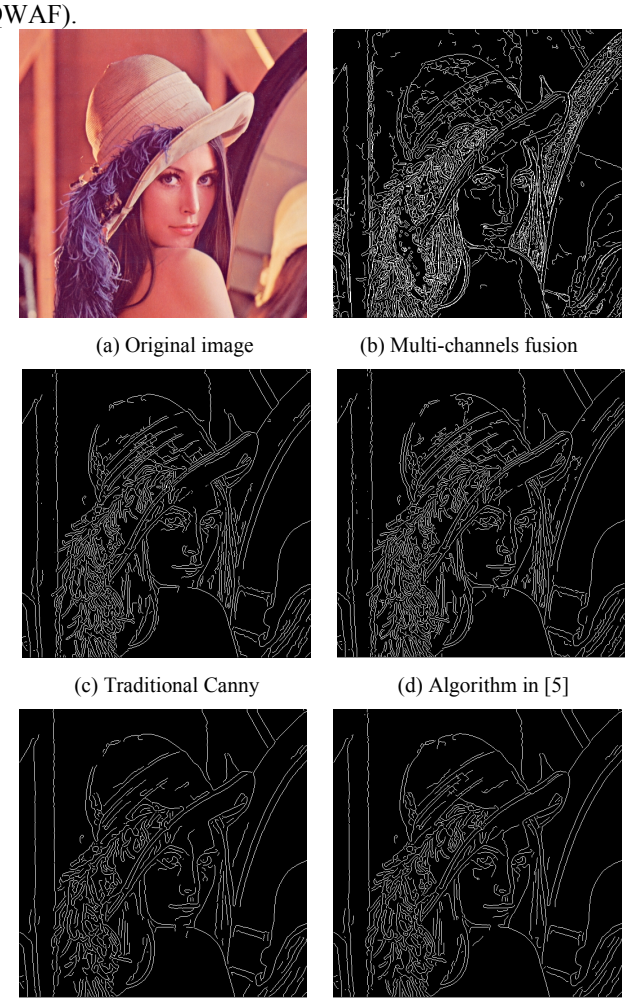
$$T_l = J_{\max} - \beta \cdot K_{\max}. \tag{31}$$

Where β is a tuning parameter and usually selected as 0.45. Define E_h and E_l are the edge images segmented by T_h and T_l separately. When the edge is discontinuous in E_h, we detect edge in E_l for complementarity.

IV. EXPERIMENT AND ANALYSIS

A. Segmentation Comparison

In this part, we use different algorithms to segment color Lena image. The algorithms used in this experiment are multi-channels fusion segmentation based on $I_1 I_2 I_3$, the traditional Canny algorithm (transfer color image to gray first), algorithm in [5] and the improved method with Gaussian filter (replace the QWAF with Gaussian filter to see the performance of QWAF).



(a) Original image (b) Multi-channels fusion

(c) Traditional Canny (d) Algorithm in [5]

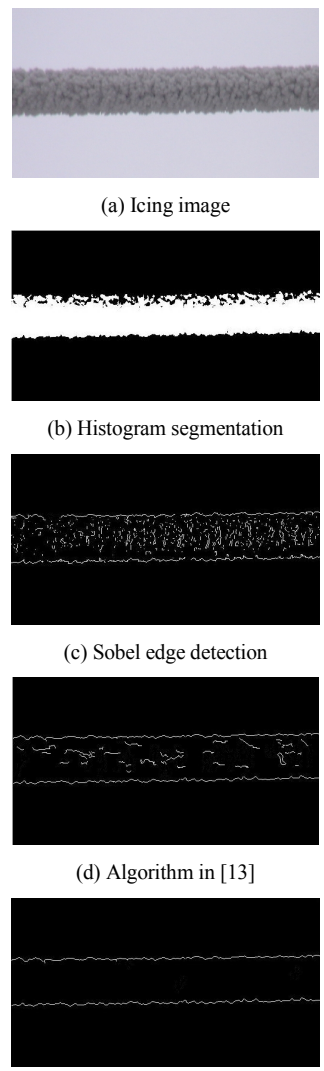
(e) Our algorithm with Gaussian filter (f) Proposed algorithm

Figure 3. Lena image segmentation comparison

From Fig. 3, we can notice that the multi-channels fusion segmentation based on $I_1 I_2 I_3$ does not have a good performance. The segmentation result contains many fake edges which will do harm to further analysis. Both the algorithms used in Fig. 3(c) and (d) need to transfer the color image to gray. The segmentation results show that thare are some discontinuous lines in Fig. 3(c) and (d), especially in the part of Lena's hat. This means that the process from color to gray leads to a information loss. Comparing Fig. 3(d) and (f), we can notice that both the two figures have a good segmentation performance. The outlines are both detected in the two figures and fake edges are fewer than other methods. However, it is necessary to notice that the outline in Fig. 3(f) is intacter than the one in Fig. 3(e). That means our proposed filter QWAF has a better performance than the traditional Gaussian filter.

B. Application

In this part, we use our proposed algorithm to deal with the image of transmission line icing. On-line monitoring system of transmission line icing based on machine vision is a new way to monitor icing disaster in the grid [11]. The theory of the monitoring system is to acquire the icing image and compute the diameter of icing line directly by analysis of the image. Image segmentation is the most important method to detect the outline of the icing line. The traditional methods for icing image, such as histogram segmentation [12], Sobel edge detection and algorithm in [13], are all gray image processing methods. As aforementioned, useful information is lost during the process when we transfer color image to gray one. Thus, the color image processing method is involved to deal with the icing image.



(a) Icing image

(b) Histogram segmentation

(c) Sobel edge detection

(d) Algorithm in [13]

(e) Proposed algorithm

Figure 4. Application comparison with gray image processing methods

The algorithms used in Fig. 4(b), (c) and (d) are all gray image processing methods used in the analysis of icing image. From Fig. 4(b), it can be seen that histogram method is difficult to divide the icing line from the background because the threshold is hard to be determined. From Fig. 4(c), we can notice that the Sobel operator can segment the object from background. While, fake edges and details are also acquired and the outline is discontinuous. Fig. 4(d) shows that the algorithm in [13] is able to detect the outline but fake details still exists. Fig. 4(e) indicates that a good segmentation result can be acquired by our proposed method. Notice that the outline is continuous and the fake edges are less. Thus our proposed color image Canny algorithm has a better performance than the gray image processing methods and can be used in the real engineering field.

V. CONCLUSION

An improved Canny algorithm based on vector concept is proposed in this paper. This algorithm can be seen as an expansibility of traditional Canny from gray to color space. Comparison results show that the proposed algorithm outperforms other algorithms for outline detection and fake edges supression. Moreover, the proposed algorithm is also applied to deal with color images of transmission line icing. The results show that our algorithm is still better than the gray image processing methods. This means that color image processing is a better way for image analysis and the proposed algorithm can be used to segment color images.

REFERENCES

[1] H.D. Cheng, X.H. Jiang, Y. Sun and Jingli Wang, "Color image segmentation: advances and prospects," Pattern Recognition, 2001, Vol. 34 No. 12, pp. 2259-2281.

[2] K. N. Plataniotis and A. N. Venetsanopoulos, Color Image Processing and Applications, Berlin: Springer-Verlag, 2000, pp. 237-273..

[3] J. Canny, "A computational approach to edge detection," IEEE Trans. Pattern Analysis and Machine Intelligence, 1986, Vol. 8 No. 6, pp. 679-698.

[4] Jianlin Liu and Xianbin Wen, "New adaptive Canny edge detection based on K neighbor mean filter," Journal of Tiangjin University of Technology, Vol. 23, No. 3, pp. 42-45.

[5] Wenliang Du Xiaodong Liu and Yuanyuan Lv, "Color image edge detection based on C-Canny operator and gray space," Microelectronic & Computers, 2000, Vol. 27, No. 4, pp. 17-20.

[6] O.N. Subakan and B.C. Vemuri, "A quaternion framework for color image smoothing and segmentation," International Journal of Computer Vision, 2011, Vol. 91, No. 3, pp. 233−250.

[7] L. Jin, H. Liu, X. Xu and E. Song, "Color impulsive noise removal based on quaternion representation and directional vector order-statistics," Signal Processing, 2011, Vol. 91, No. 5, pp. 1249−1261.

[8] P. Denis, P. Carre, and C. Fernandez-Maloigne, "Spatial and spectralquaternionic approaches for colour images," Computer Vision andImage Understanding, 2007, Vol. 107, No. 1/2, pp. 74−87.

[9] Xin Geng, Xiaoguang Hu and Jin Xiao, "Quaternion swithcing filter for impulse noise reduction in color image," Signal Processing, 2012, Vol. 92, No. 1, pp. 150-162.

[10] Yujin Zhang, Image Engineering (II): Image Analysis, 2rd ed., Beijing: Tsinghua University Press, 2005, 73-85.

[11] Xin Geng and Xiaoguang Hu, "On-line Monitoring Method and Observation System of Transmission Line Icing," ICIEA 2009, Xi'an, China, 2009, pp. 1265-1269.

[12] S. M. Berlijn, "Laboratory tests and web based surveillance to determine the ice and snow performance of insulators," IEEE Transactions on Dielectrics and Electrical Insulation, 2007, Vol. 14, No. 6, pp. 1373-1380.

[13] Xiaopeng Wang and Jianlin Hu, "Study on edge extraction methods for image-based icing on-line monitoring on overhead transmission lines," High Voltage Engineering, 2008, Vol. 34, No. 12, pp. 2687-2693.